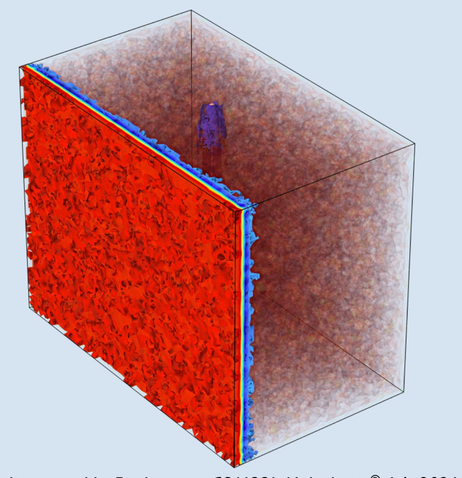

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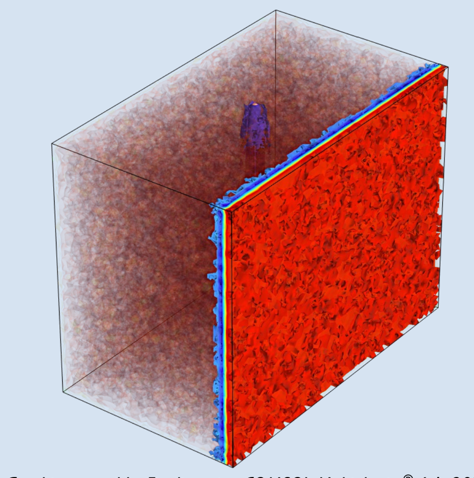
Curing Cancer in a FLASH: A Finite-Element Computational Analysis of an Ultrasonic Detection Array Setup for Ultra-High Dose Rate (FLASH) Measurements and Absolute Dosimetry



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INTRODUCTION

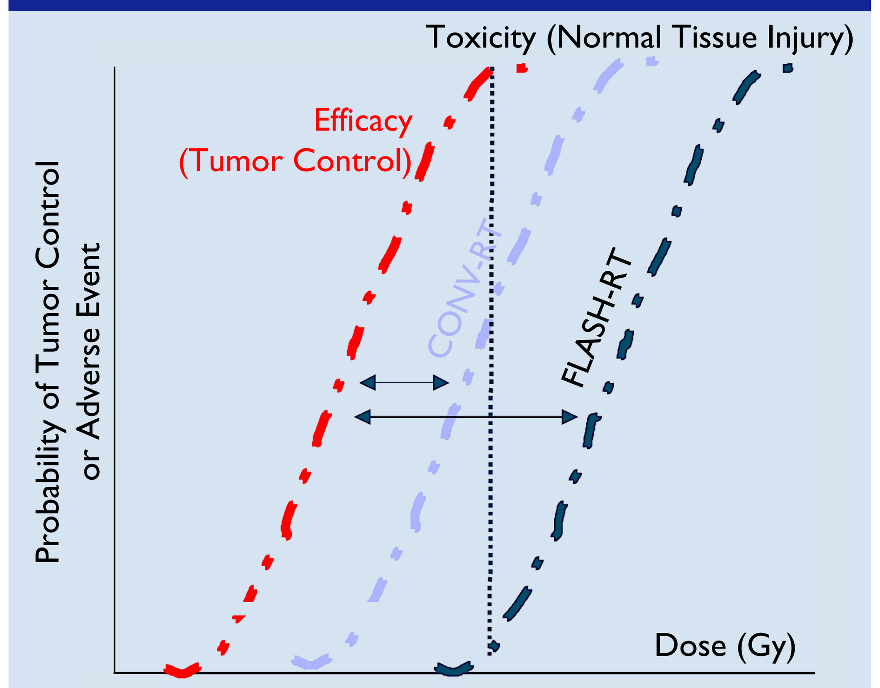

Figure 1: Therapeutic Window for FLASH-RT and CONV-RT *[1]*

Conventional Radiotherapy (CONV-RT)
- 0.01 – 0.40 Gy/s dose rates
- Minute-long pulses
- Relatively many fractions
- Irradiation damage to healthy tissue

FLASH Radiotherapy (FLASH-RT)
- Ultra-high dose rate (≥ 40 Gy/s)
- Under 200 ms
- Relatively few fractions
- **Minimizes irradiation damage to healthy tissue**

SHORTCOMINGS OF FLASH-RT

Problem: Clinical Adoption
- Need to precisely verify dose rate within a volume
- **Not possible** with **current dosimetry hardware** or **measurement standards**

PURPOSE

This work seeks to address **both shortcomings** by developing an **absolute dose** and **dose rate detection principle** based on **Phase Modulation** (measurement standard) of **ultrasound** (hardware). Specifically, by a Finite-Element Method (FEM) analysis of the proposed detection setup via simulations of a completely thermoacoustic model on the COMSOL Multiphysics® 6.1 software.

METHODOLOGY

Phase 1: Theoretical Framework
Justify the use of Phase Modulation over traditional Time-of-Flight to reconstruct delivered dose to a tumor cell.

Phase 2: Computational Framework
Design a completely thermoacoustic water phantom experiencing FLASH-level radiation doses and orthogonally-propagated, continuous ultrasound

Analysis and Comparison
Conduct Finite-Element-Method (FEM) simulations of Acoustic Pressure, comparing theoretic yields to the computed phase-dilation factor

Figure 2: Methodology Workflow *[2]*

THEORETICAL MOTIVATIONS

Positively Temperature-Dependent Speed of Sound

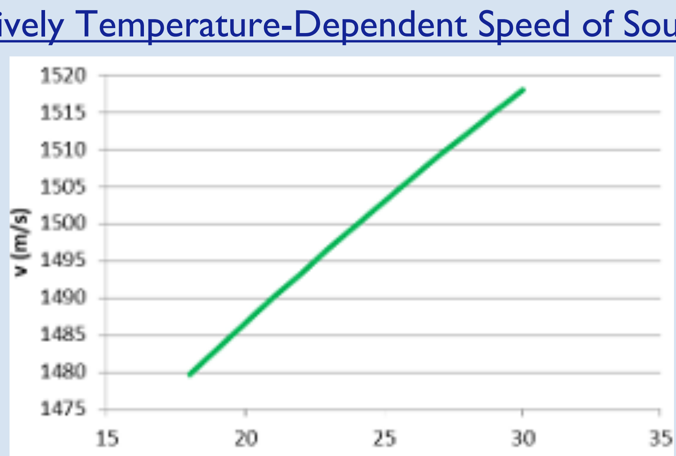

Figure 3: Speed of Sound in Water vs. Temperature *[3]*

Time-Dependent Phase Shift from Radiation Heating

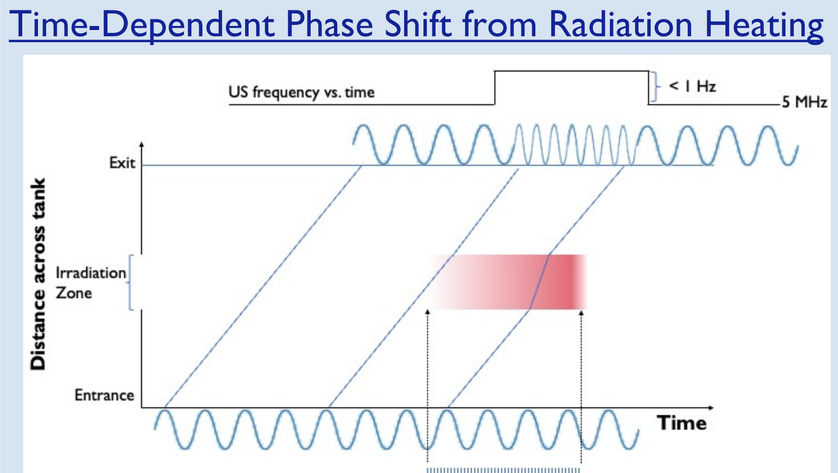

Figure 4: Ultrasonic Frequency vs. Time across Undisturbed and Irradiated Regions *[4]*
- Periodic application of radiation frequency-modulates the injected carrier of 5 MHz

PHASE 1: THEORETICAL FRAMEWORK

Temperature-Dependent Speed of Sound

$$\boxed{v(T) \cong 1394 + 4.5T}$$

Induced Heating from Delivered Dose

$$\boxed{\Delta T \cong 0.2391 \text{ mK/Gy}}$$

Traditional Time-of-Flight (ToF)

$$\boxed{\Delta ToF \cong 10 \frac{\text{ps}}{\text{Gy}}}$$

$$\boxed{\Rightarrow ToF \cong 7\ \mu s \text{ per cm } (\sim 2 \text{ cm})}$$

- **Difficult** to measure experimentally
- Requires **expensive** (> $10,000) and **specialized** hardware (Stanford Research Systems or Zurich Instruments Lock-In Amplifiers)

Phase Modulation

$$\boxed{\Delta\varphi = \Delta(\omega \cdot ToF)}$$

$$\boxed{\cong \omega \cdot \Delta ToF \cong 18 \frac{\text{mdeg}}{\text{Gy}}}$$

- **Easily-observable** changes in experiment
- Can be measured with **low-cost** and **general** technologies (RF Spectrum Analyzers or Amateur Radio-based detectors)

PROPOSED SETUP

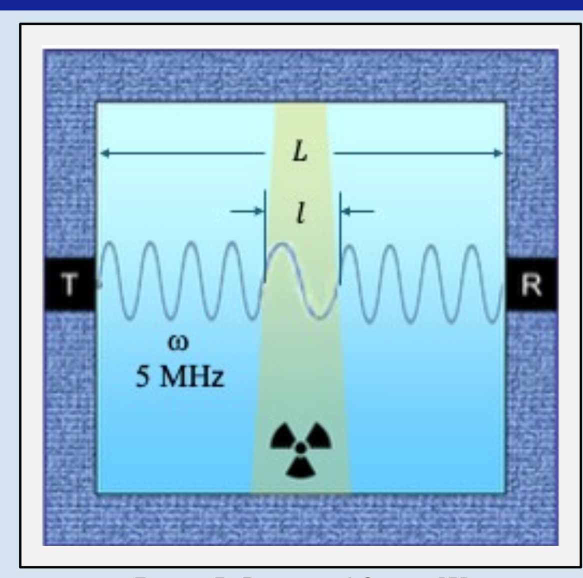

Figure 5: Proposed Setup *[5]*

Propagate an ultrasonic wave orthogonal to a radiative beam in a water phantom to observe potential phase modulation of the ultrasound for de-modulation, determining the delivered dose and dose rate.

THERMAL MODEL VALIDATION

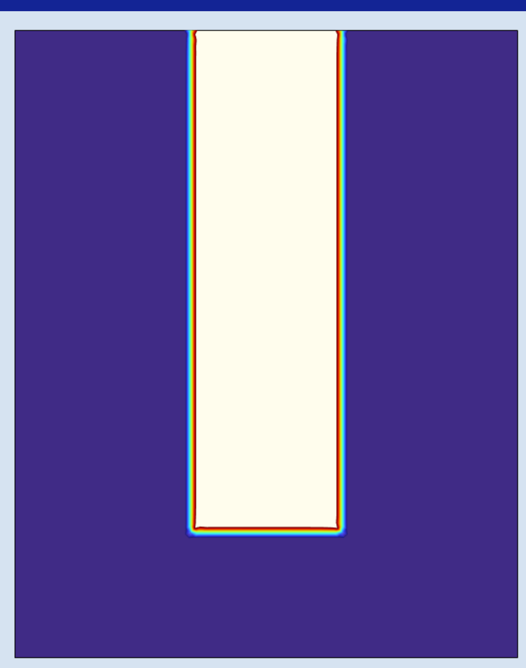

Figure 6: Sub-Kelvin Heat Contours from a FLASH-RT Radiative Beam *[6]*
- Temperature surface plot, with a vertical radiative beam, modeled by a Heat Flux condition coupled with the Radiative Beam sub-module
- Sub-Kelvin heating of a FLASH-RT dose **validated**
  - Approximately 10 mK temperature difference achieved by a 40 Gy/s dose rate beam

ACOUSTIC MODEL VALIDATION

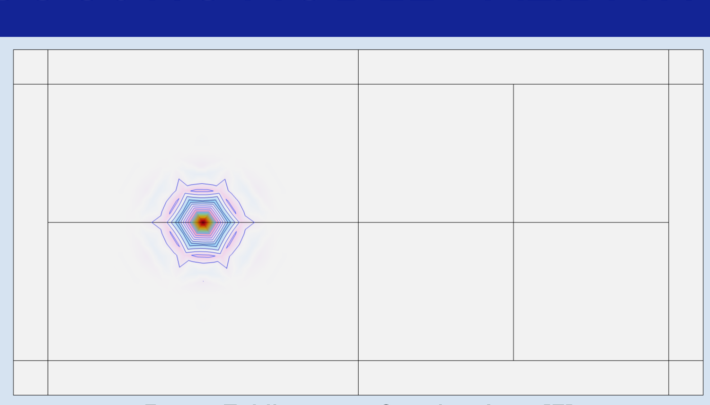

Figure 7: Ultrasonic Signal at 1 μs *[7]*
- Modified High-Intensity Focused Ultrasound (HIFU) macro, following experimental dimensions

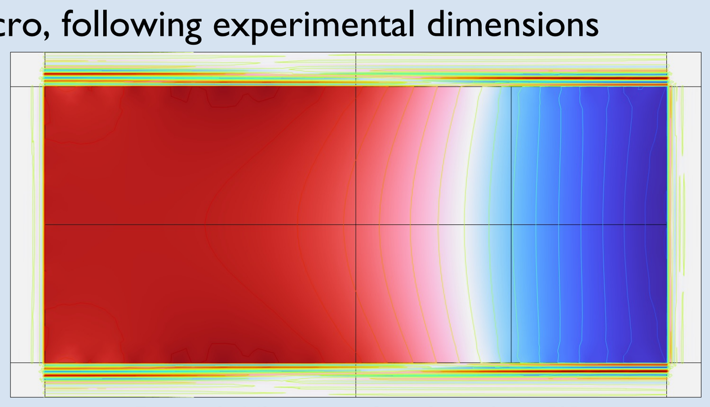

Figure 8: Ultrasonic Signal at 60 μs *[8]*
- Localization of Normalized Acoustic Pressure in the less-dense heated region **validated**

THERMOACOUSTIC THEORY

Heat Transfer with Radiative Beam in Absorbing Media
- Intensity Attenuation (Beer-Lambert Law):

$$\boxed{\nabla \cdot (I_i \vec{e_i}) = -\kappa I_i}$$

- Heat Transfer Equation (Radiation Energy Deposition):

$$\boxed{\rho C_p \frac{\partial T}{\partial t} + \rho C_p \vec{u} \cdot \nabla T + \nabla \cdot \vec{q} = Q + \sum_i \kappa I_i}$$

Pressure Acoustics
- 2-dimensional axisymmetric cross-section:

$$\boxed{p(r, \varphi, z) = p(r, z) e^{-im\varphi}}$$

- Nonreflecting Boundary Condition approximation:

$$\boxed{\Delta = \frac{1}{\omega^2}}$$

Coupled Frequency (Time-Harmonic) Domain
- Inhomogenous Helmholtz Equation:

$$\boxed{\nabla \cdot \left(-\frac{1}{\rho_c} (\nabla p_{\text{total}} - \vec{q_d}) \right) - \frac{k_{\text{eq}}^2 p_{\text{total}}}{\rho_c} = Q_{\text{m}}}$$

Wave Propagation
- 5 MHz ultrasonic waveform propagated in z-direction to model standard transducers used in experiment

PHASE 2: COMPUTATIONAL FRAMEWORK

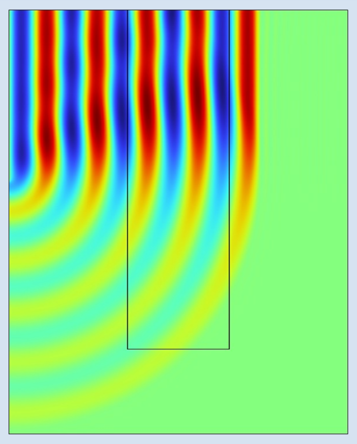

Figure 9: Surface Plot of Total Acoustic Pressure at 1 μs *[9]*

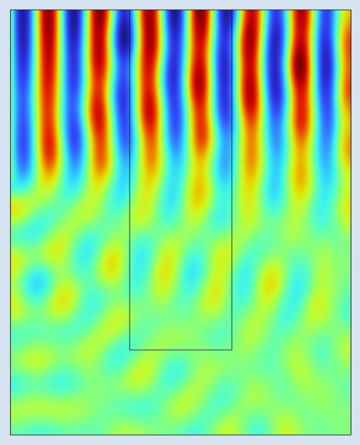

Figure 10: Surface Plot of Total Acoustic Pressure at 4 μs *[10]*

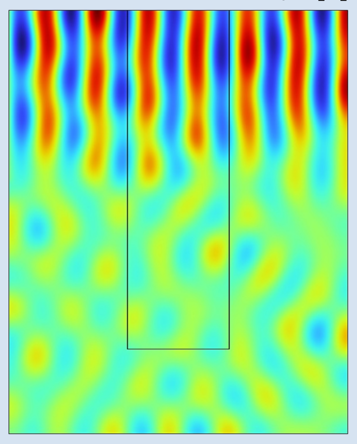

Figure 11: Surface Plot of Total Acoustic Pressure at 47 μs *[11]*

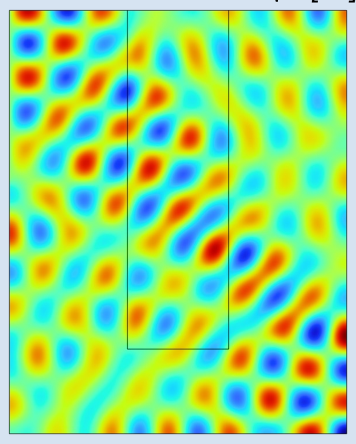

Figure 12: Surface Plot of Total Acoustic Pressure at 60 μs *[12]*

ACOUSTIC PRESSURE ANALYSIS

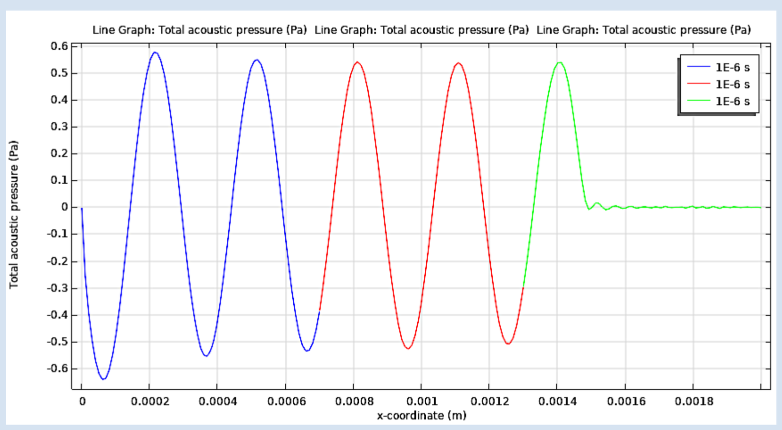

Figure 13: Line Plot of Total Acoustic Pressure at 1 μs *[13]*

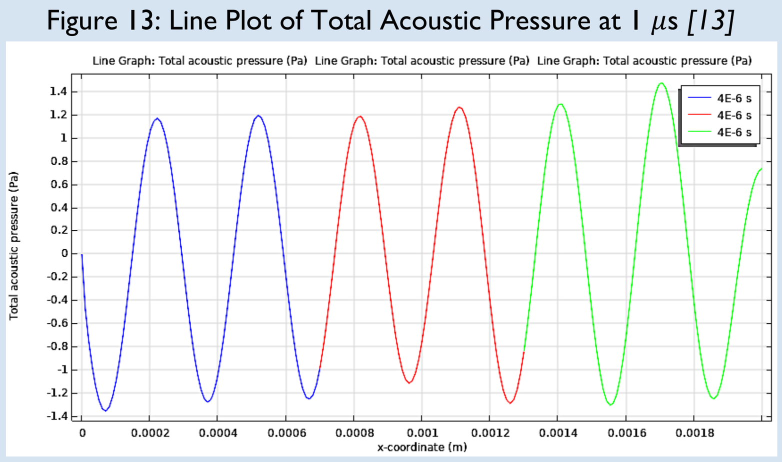

Figure 14: Line Plot of Total Acoustic Pressure at 4 μs *[14]*

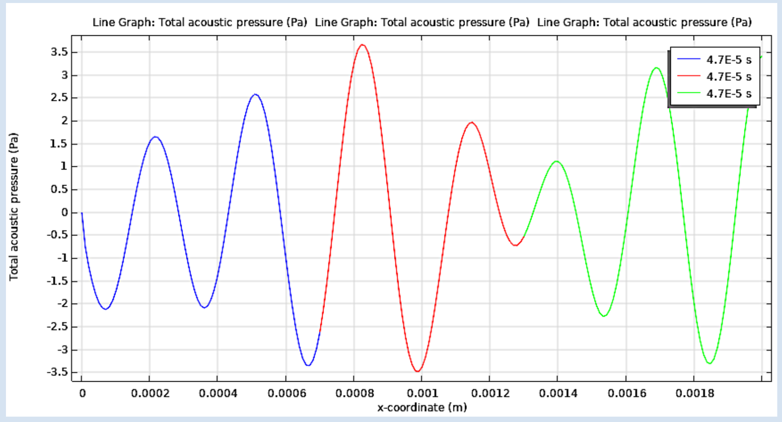

Figure 15: Line Plot of Total Acoustic Pressure at 47 μs *[15]*

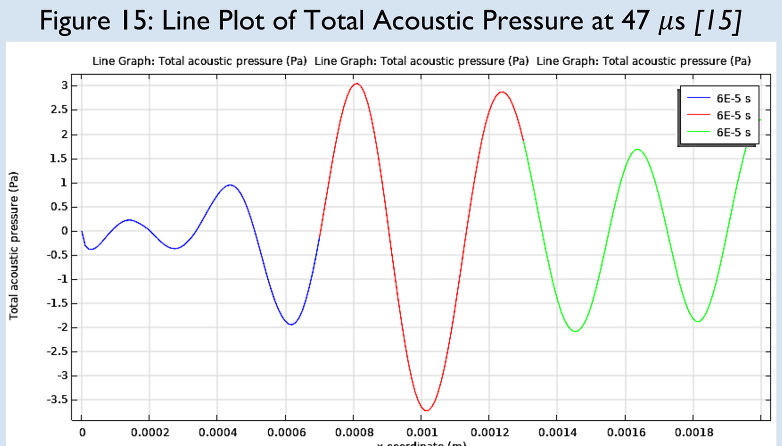

Figure 16: Line Plot of Total Acoustic Pressure at 60 μs *[16]*

PHASE SHIFT: A COMPARISON

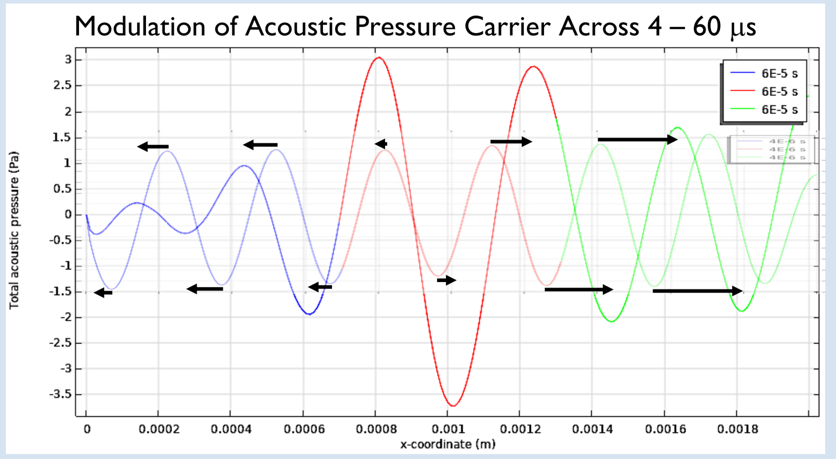

Figure 17: Total Acoustic Pressure Comparison *[17]*

Phase Dilation Factor: Pre- vs. Post- Irradiation
- **Theoretical Yield of 3.0** comparable to a **Simulated Yield of ≈2.9**

ERROR ANALYSIS

- Computational Power shortage
  - 'Extremely Fine' density (> 200,000 elements), yet static mesh throughout study steps

SUMMARY OF RESULTS

Phase Modulation: A Superior Measurement
- The scale of changes in phase is **9 orders of magnitude** greater than the scale of changes in Time-of-Flight
- Can be easily measured with non-specific equipment already available in clinics

Successful Detection of Delivered Dose
- Significant Phase Modulation of Ultrasonic Signal
- Slight Amplitude Modulation
  - Close agreement with preliminary results observed in experiment using a reconfigured RF-powered, traveling-wave electron linac

EXPERIMENTAL SETUP

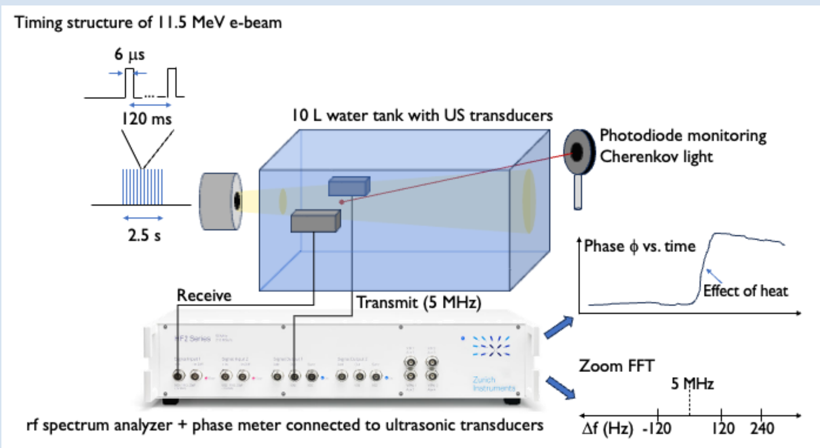

Figure 18: Experimental Setup *[18]*

CONCLUSION

Problem Revisited: Clinical Adoption
- Need to precisely verify dose rate within a volume
- **Not possible** with **current dosimetry hardware** or **measurement standards**

Solution: Novel Ultrasonic Detection Array Setup
- **Hardware:** use of Ultrasound
- **Measurement Standard:** Phase Modulation
  - $\Delta\varphi = \Delta(\omega \cdot ToF) \cong \omega \cdot \Delta ToF \cong 18 \frac{\text{mdeg}}{\text{Gy}}$

APPLICATIONS

General Radiation Oncology
- Enable **low-cost**, **precise** verification of **delivered radiation dose** to a tumor
- Establish a new **primary standard** for **Absolute Dosimetry**

FLASH Radiotherapy
- Provide **patient** and **insurance guarantees** on treatments
- Further FLASH-RT's status into a **clinical modality**

ACKNOWLEDGMENTS

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GRAPHIC REFERENCE LIST

[1] Cengel, K. A., Kim, M. M., Diffenderfer, E. S., & Busch, T. M. (2024). FLASH Radiotherapy: What Can FLASH's Ultra High Dose Rate Offer to the Treatment of Patients With Sarcoma? *Seminars in Radiation Oncology*, *34*(2), 218–228. https://doi.org/10.1016/j.semradonc.2024.02.001.
[2] *Chart created by Finalist, 2026.*
[3] *Graphic created by Finalist using Microsoft® Excel®, 2026.*
[4, 5] *Graphic created by Finalist using Microsoft® PowerPoint®, 2026.*
[6-16] *Graphic created by Finalist using COMSOL Multiphysics® 6.1, 2026.*
[17, 18] *Graphic created by Finalist using Microsoft® PowerPoint®, 2026.*